

Lily Hoffman

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EDUCATION

Northeastern University, Boston, MA

Apr. 2026

Bachelor of Science in Data Science and Psychology

Relevant Coursework: Large-Scale Storage and Retrieval | Advanced Programming with Data | Database Design | Machine Learning and Data Mining II | Information Visualization

GPA: 3.9/4.0 | **Honors:** Summa Cum Laude | University Honors Program | Dean's List

TECHNICAL SKILLS

Languages: Python | SQL | Scala | Bash

Data Engineering: Apache Spark (PySpark) | Kafka | Airflow | dbt | Databricks | ETL/ELT | Data Modeling

Cloud & Infrastructure: AWS (S3, Glue, Athena, EC2, Lambda, RDS) | Terraform | Docker

Databases & Tools: PostgreSQL | BigQuery | MySQL | MongoDB | Redis | Git

EXPERIENCE

Khoury College of Computer Sciences, Boston, MA

Undergraduate Teaching Assistant

Sep. 2024 – May 2026

- Resolved SQL and ETL issues for ~20 students weekly by debugging query logic and job configuration.
- Diagnosed recurring environment and setup failures (e.g., dependencies, database connections) across 100+ students per semester, sharing fixes in office hours to reduce repeat issues.
- Advised 4–5 teams per semester on pipeline and data model design across relational and NoSQL stores (Redis, Cassandra); graded weekly Python and SQL homework.

Aura, Boston, MA

Data Analyst Co-op

Jul. 2025 – Dec. 2025

- Automated customer dissatisfaction root-cause reporting with a Scala/SQL pipeline in Databricks, saving 10 customer experience (CX) users a combined ~3 hours of manual pulls weekly.
- Collaborated with Data Engineering to build a daily Databricks job that auto-populates 10 weekly business review charts with 6-week trend metrics, replacing 3 hours/week of manual updates.
- Cut runtime of a CTE-based SQL query behind a CX KPI dashboard by ~40% (12 to 7 minutes) by replacing UNION with UNION ALL and removing sorts.

UKG, Lowell, MA

Data Science Practice Intern

Oct. 2024 – Apr. 2025

- Enabled no-code prompt updates for an internal AI chatbot by replacing hard-coded logic with configuration-driven Python in a Plotly Dash app used by 75% of the business development team.

Marketing Operations Intern

May 2024 – Sep. 2024

- Reduced weekly processing time for 500+ EMEA unsubscribe URLs by ~97% with a Python and Selenium automation that replaced manual browser work.
- Streamlined 30+ recurring data workflows with VBA and XML automation, eliminating 3 hours/week of manual data processing.

DATA ENGINEERING PROJECTS

Financial Markets Data Platform – Python, PySpark, PostgreSQL, dbt, Airflow

Sep. 2026

- Built a containerized batch pipeline for 5 tickers with incremental Yahoo Finance ingestion into a Bronze/Silver/Gold architecture, using PySpark for schema enforcement, deduplication, and price/volume validation.
- Implemented transactional PostgreSQL loads and dbt tests in a weekday Airflow DAG with retries and audit logging; reruns insert zero duplicate rows.

Real-Time E-Commerce Streaming Pipeline – Kafka, PySpark, PostgreSQL, Docker

Sep. 2026

- Engineered a Kafka and PySpark Structured Streaming pipeline that validates simulated orders and computes 5-minute event-time revenue and order metrics by category, with a 10-minute watermark for late data.
- Implemented idempotent PostgreSQL writes and Spark checkpointing, verifying the job resumed after a mid-stream restart and processed events published during downtime.

Cloud Analytics Data Platform – PySpark, AWS S3, AWS Glue, Athena, Terraform

Sep. 2026

- Designed a serverless AWS data lake, provisioned with Terraform, that processes 3.48M NYC taxi records into partitioned Parquet datasets queryable in Athena.
- Improved data reliability by filtering 151K+ invalid records (about 4% of the dataset) with PySpark quality checks, and cataloged the cleaned data with Glue.